

MARVIN WONG

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EDUCATION

University of California, Los Angeles (UCLA) | Los Angeles, CA

Sep 2025 - Dec 2026

Master of Engineering, Data Science

- **GPA:** 4.0
- **Coursework:** Neural Networks and Deep Learning, Natural Language Processing, Large-Scale Machine Learning, Product Management, Financial Management

University of California, Los Angeles (UCLA) | Los Angeles, CA

Sep 2021 - Jun 2025

Bachelors of Science - Data Theory, Atmospheric and Oceanic Sciences

- **GPA:** 3.832
- **Coursework:** Statistical Modeling, Advanced Linear Algebra, Programming with R, Data Analytics, Statistical Models in Finance, Design and Analysis of Experiment, Machine Learning, Probability, Data Structures and Algorithms

TECHNICAL SKILLS

- **Skills:** Python, Tableau, SQL, Pandas, Matplotlib, R, Microsoft Office, Sci-kit Learn, PyTorch, Prompt Engineering, Data Visualization, Machine Learning, Predictive Modeling, AWS, Spark, Databricks, ETL, TensorFlow, Excel, Numpy, Docker, MCP

EXPERIENCE

UCLA Computer Science Department

Mar 2026 - Present

Graduate Research Assistant

- Designed an AI-driven data retrieval system that optimized matching infrastructure, utilizing advanced geometric analysis (optimal transport) to **increase model success rates from 70% to 90%**
- Engineered an automated **Machine Learning** preprocessing workflow using vision-based **embedding models** and **PyTorch** to segment and organize large-scale datasets for downstream training

Hong Kong University of Science and Technology | Hong Kong

Jun 2024 - Sep 2024

Research Assistant - Atmospheric Data Scientist

- Built an automated data ingestion pipeline in Python that scraped and consolidated over 3 GB of NOAA climate data, using Pandas for cleaning and Scikit-learn for preprocessing, which streamlined data preparation and enabled timely analysis for the research team
- Developed and optimized a **predictive time-series forecasting model** (TDA) for climate signals, **improving model accuracy by 50%** compared to baseline methods
- Evaluated distribution similarities using Wasserstein distances, leveraging **Matplotlib** to engineer granular **data visualizations** of model performance metrics (**RMSE**) and high-dimensional feature distributions

PricewaterhouseCoopers | Hong Kong

Jul 2023 - Sep 2023

Data Analyst Intern

- Developed executive-level Tableau visualizations and reports on key performance indicators and conversion metrics, delivering data storytelling that translated complex data into actionable business insights for client stakeholders
- Revamped UI/UX for legacy SAP procurement system, increasing user satisfaction by 15% and facilitating 15% more transactions handled per day
- Executed A/B Testing experiments on 5 distinct UI features, applying statistical inference to determine significance and optimize features, lifting user engagement by 15%
- Conducted 4 company-wide training sessions that led to 80% switchover rate to new system

Hysan Development Company

Jun 2022 - Sep 2022

Intern

- Conducted market research for a proposed skate park in Hysan's portfolio and modeled foot traffic and revenue scenarios in Excel, providing data that helped senior management approve the strategic development proposal
- Engineered a stock ledger system using VLOOKUP for automated data validation and Pivot Tables for dynamic reporting, reducing manual entry errors and cutting reconciliation time by 2 hours per week

ASUCLA

Feb 2023 - Jun 2025

Student Supervisor

- Designed and executed a standardized training framework for 10 new hires, accelerating their time-to-autonomy and ensuring consistent adherence to quality and safety standards
- Labeled and monitored peak-hour service bottlenecks; reconfigured team deployment and physical floor layout to increase shift throughput by 15% during high-volume periods
- Audited and reconciled daily financial logs, cash drawers, and safe drops with zero variance, ensuring high data accuracy for corporate accounting reviews

PROJECTS

Securiport Visual Intelligence Capstone

Jun 2026 - Present

Securiport

- Engineered an end-to-end multimodal **Machine Learning pipeline** in **Python** and **PyTorch**, deploying state-of-the-art **Computer Vision** models (YOLOv8 and DINOv2) to perform automated object detection and high-dimensional **feature extraction**
- Designed and built a **high-performance search engine** using **Scikit-learn** and **NumPy** to compute cosine similarity metrics, integrating object tracking algorithms to maintain target classification data across live video streams
- Deployed **Large Language Models (LLMs)** and Vision-Language architectures to automate behavioral classification, processing sequential data frames and metadata into structured time-series summaries

AI Football Scouting Platform

- Built an automated data ingestion, transformation and cleaning pipeline in **Python** to consolidate player metrics, salaries, market values and crowd-sourced ratings into a central **SQL** database, enabling instant multi-condition filtering to discover top talent
- Developed an AI scouting assistant using **LangChain** and **Google Gemini**, leveraging a **vector database with Retrieval-Augmented Generation (RAG)** to search written narrative scouting reports (e.g., personality traits, injury history) with an automated web search backup for missing data
- Engineered dynamic visualization tools within an interactive **Streamlit** dashboard that allow the **AI agent** to automatically generate visual player percentile comparison charts on demand based on user queries
- Optimized system performance and API costs by writing a background cleanup routine that stripped heavy chart image data from the AI's memory, preventing app crashes while maintaining seamless conversational memory

Amazon Multimodal Graph Recommender Model

- Engineered a high-performance multimodal graph recommender system on Amazon Fashion, processing and vectorizing complex features (product description, user reviews, and user ratings) stored in S3 using Apache Spark for massive-scale ETL
- Architected the end-to-end MLOps pipeline on serverless AWS, utilizing SageMaker for LightGCN model training and real-time, low-latency prediction serving
- Enhanced user discovery metrics and cross-selling efficacy by executing rigorous **model evaluation** to continuously improve recommendation quality

Reddit r/AITA Language Model

- Fine-tuned a **RoBERTa** model using **PyTorch**, **Hugging Face**, and **Scikit-Learn** on 110k+ unstructured social media posts to classify complex interpersonal conflict, boosting **Macro F1-score from 0.46 to 0.71+** (a 54% gain over naive TF-IDF baselines)
- Engineered engagement and structural metadata features (e.g., comment-to-score ratios, post edits), utilizing **point-biserial correlations** and **Chi-squared tests** to uncover key behavioral signals driving community verdicts
- Mitigated severe class imbalance by implementing custom loss functions (**Focal Loss**) and threshold tuning, using **confusion matrix error analysis** to optimize precision-recall trade-offs
- Conducted quantitative feature importance and lexical coefficient analysis in **Scikit-Learn** to isolate predictive signals from linguistic noise and social context
- Packaged fine-tuned transformer models into a high-throughput **FastAPI** REST API, using **Docker** for containerization and reproducible deployment on **GCP Cloud Run**

Financial Asset Allocation and Technical Analysis Application

- Integrated a full-stack portfolio optimization web application using Python Flask reducing manual analysis time by 85%
- Engineered financial data processing and visualization systems leveraging yFinance, Pandas, and Matplotlib to process 50,000+ data points and create interactive dashboards showing risk/return metrics
- Designed responsive front-end interfaces with intuitive presentation of data and analysis results using modern CSS/HTML

BirdCLEF+ Audio Classifier | Los Angeles, CA

- Developed a streamlined preprocessing pipeline, standardizing audio data and implementing amplitude normalization and targeted noise-reduction techniques tailored for rainforest environments
- Applied pseudo-labeling techniques on unlabeled audio data to mitigate domain shifts, improving model generalization to diverse environmental conditions in passive acoustic monitoring
- Achieved final test accuracy of 0.86 on hidden test set, placing in top 10% of all 2000 submissions to competition and contributing to conservation efforts in Middle Magdalena Valley

Earthquake Data ETL Pipeline

Feb 2025 - Apr 2025

- Implemented an end-to-end ETL pipeline on Microsoft Azure to process global seismic data using Databricks and PySpark
- Ingested raw JSON from the USGS API and transformed through a Medallion architecture, enabling cleansed, query-ready datasets for advanced geospatial analytics

Football Player Clustering and Mapping

- Developed a data pipeline in Python to ingest, clean, and consolidate event data from 7 major football competitions, creating a query-ready dataset with 15+ custom key performance metrics for player evaluation
- Performed cluster analysis to segment players into distinct archetypes based on their on-field performance style, identifying the optimal number of clusters through statistical methods
- Created interactive network visualizations to map player clusters, uncovering nuanced relationships between player styles that were not apparent through traditional analysis

Stateful Log Observability AI Agent

- Implemented a stateful memory component within an **NLP AI Agent framework (LangChain)** to track log frequencies and system anomalies, automating alert escalations to optimize infrastructure reliability
- Built a **Retrieval-Augmented Generation (RAG) pipeline** using **FAISS** and **Gemini** to index text documents, successfully accelerating incident response times and **reducing mean-time-to-resolution (MTTR)**

Portfolio and Stock Analysis Application

- Developed a full-stack portfolio optimization web application using Python Flask, analyzes investment strategies and generates optimized asset allocations, reducing manual analysis time by 85%
- Engineered financial data processing and visualization systems leveraging yFinance API, Pandas, and Matplotlib to process 50,000+ data points and create interactive dashboards showing risk/return metrics
- Implemented quantitative portfolio optimization algorithms including Minimum Variance, Tangency Portfolio, and Monte Carlo simulation enhanced risk-adjusted returns by 22% in backtesting
- Designed responsive front-end interfaces with intuitive presentation of data and analysis results using modern CSS/HTML

ACCOMPLISHMENTS

- **2023 American Statistical Association DataFest Finalist:** Top 12 finalist out of 100 teams, conducted NLP sentiment analysis, statistical regression on American Bar Association appointment transcripts in Python to address customer concerns and backlog